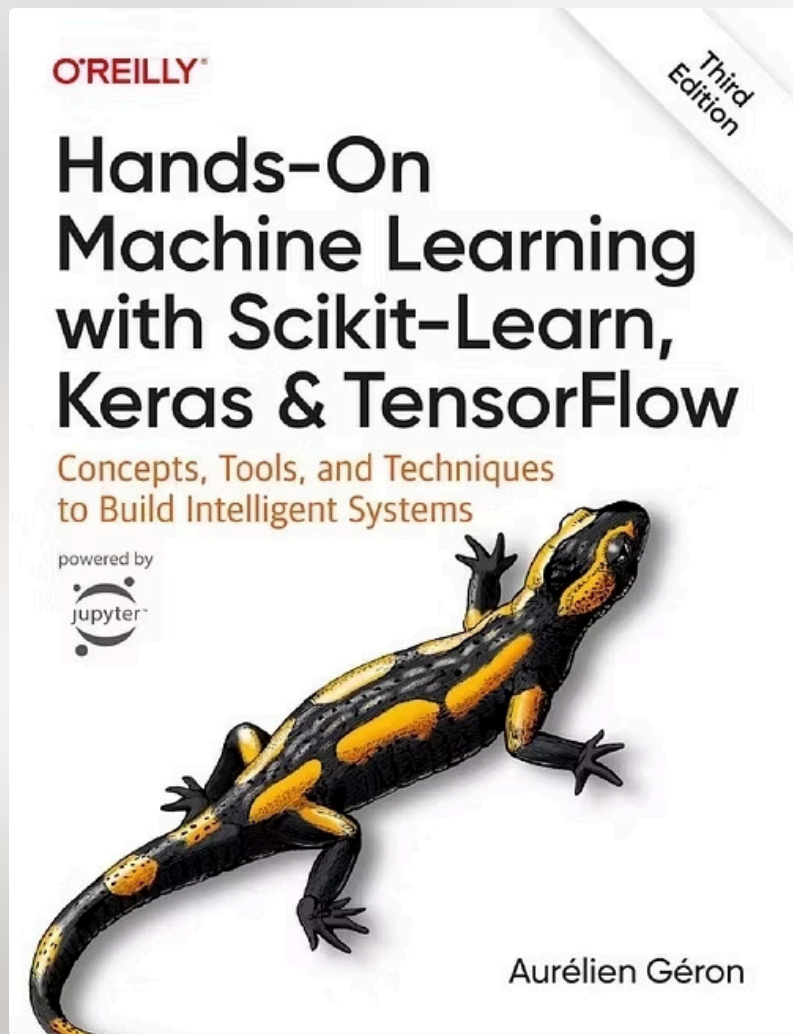




Machine Learning

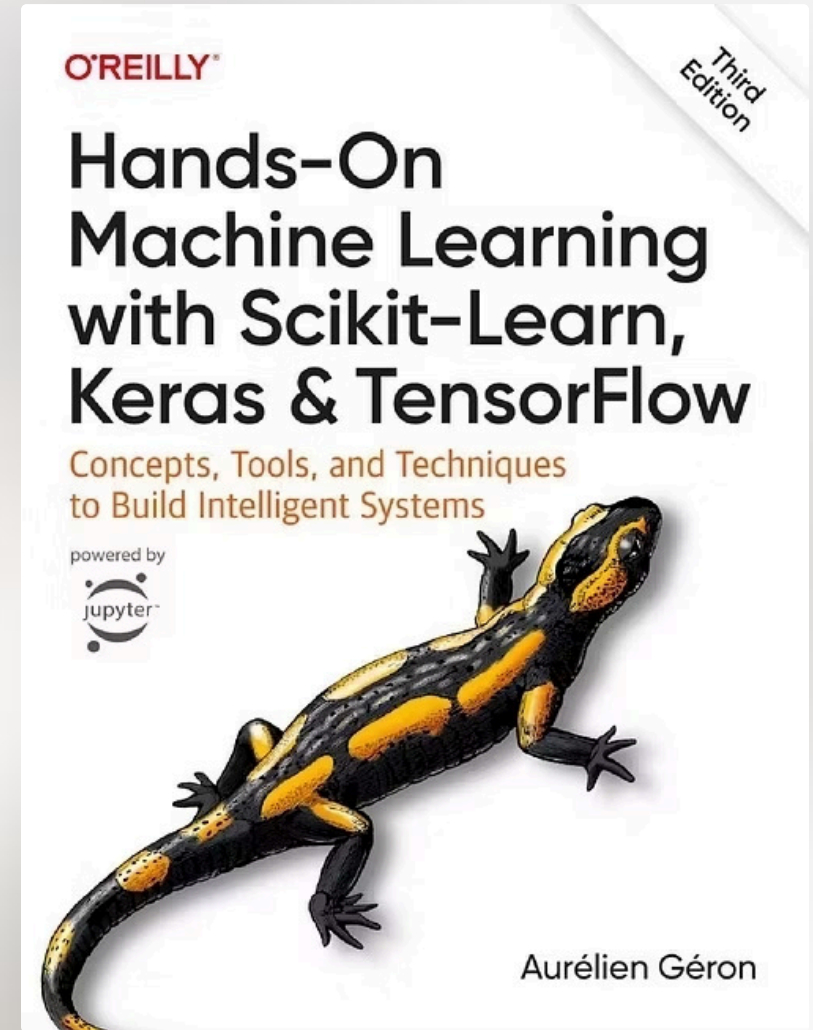
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

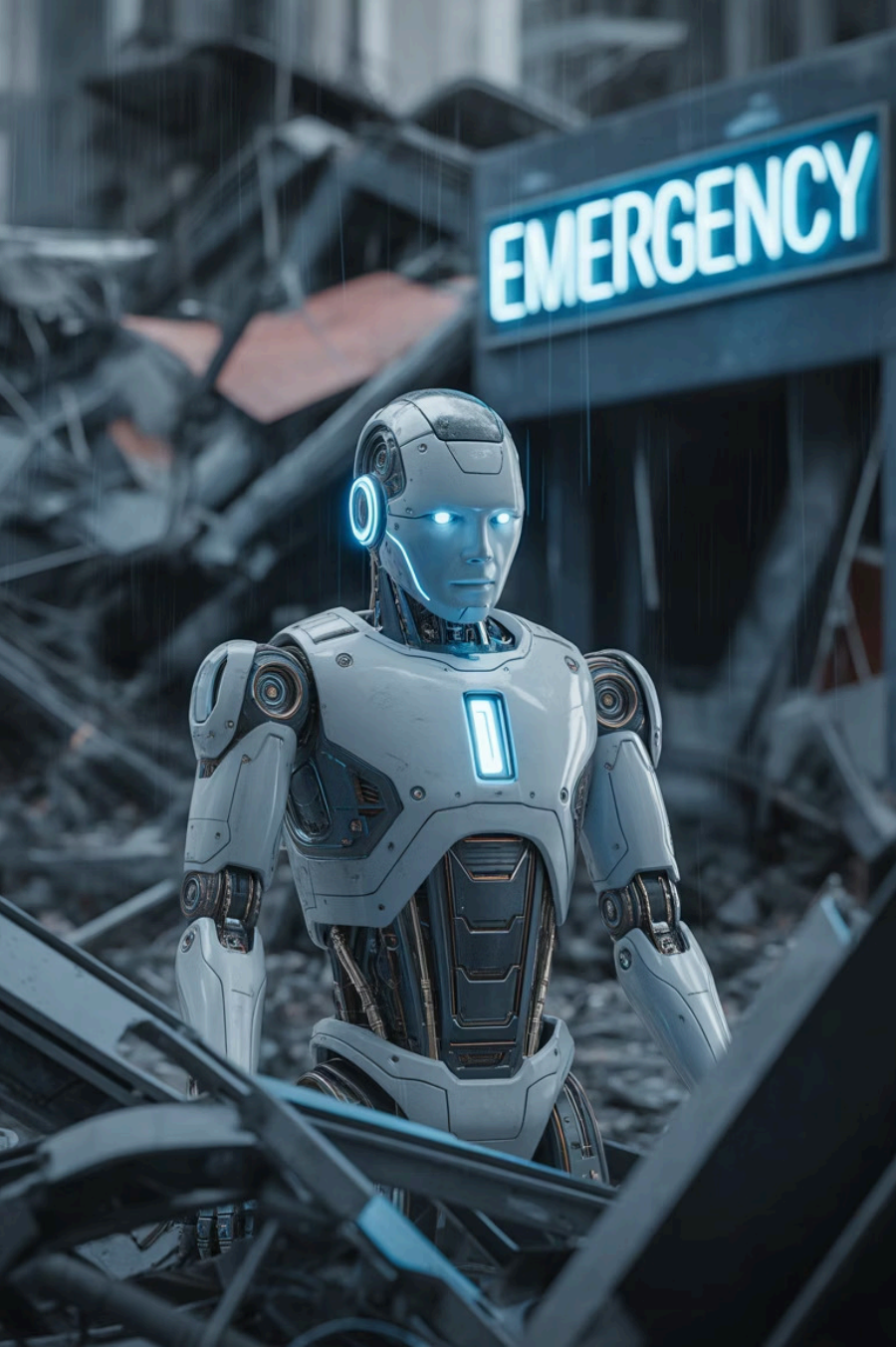
RP by Rodrigo Martins Pagliares

Last update: 08/07/2025

Machine Learning

Chapter 1 – The Machine Learning Landscape



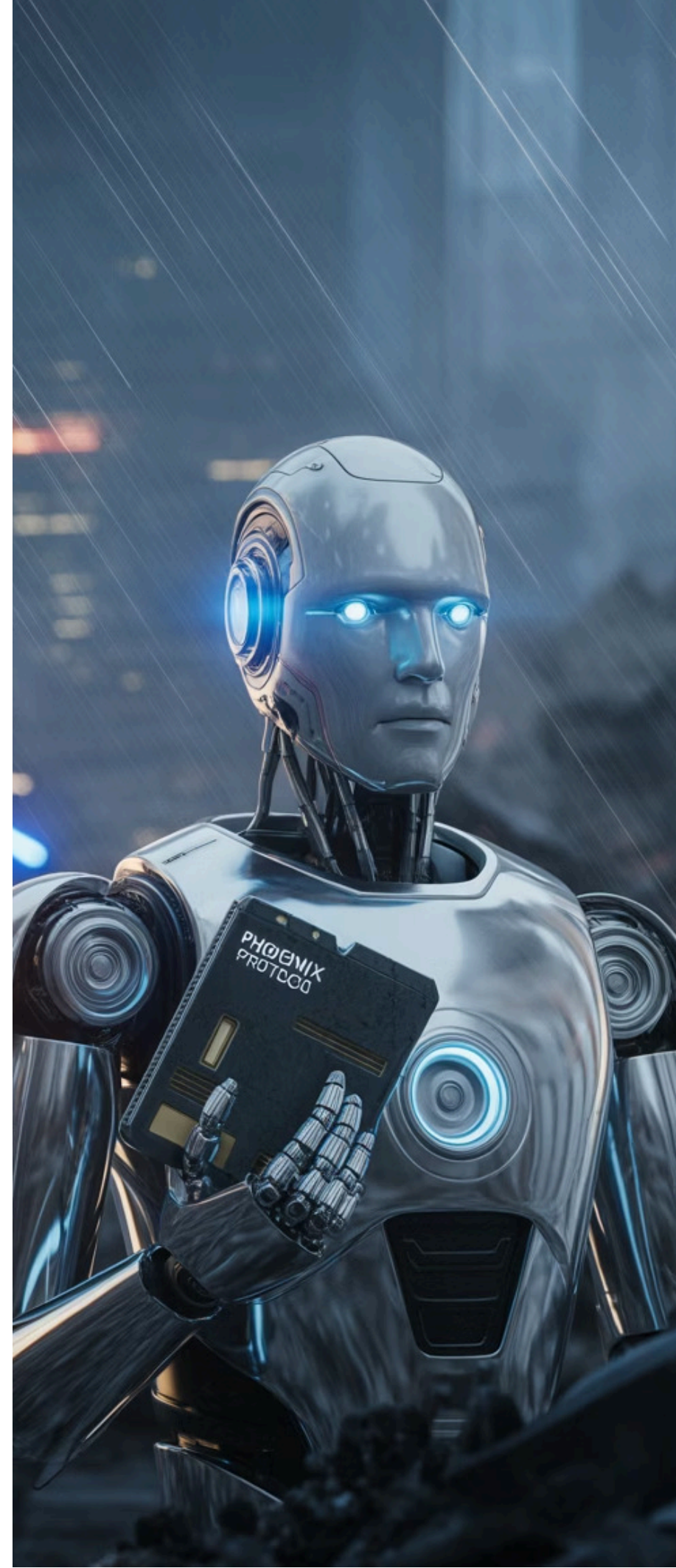


Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - **Non-representative Training Data**
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



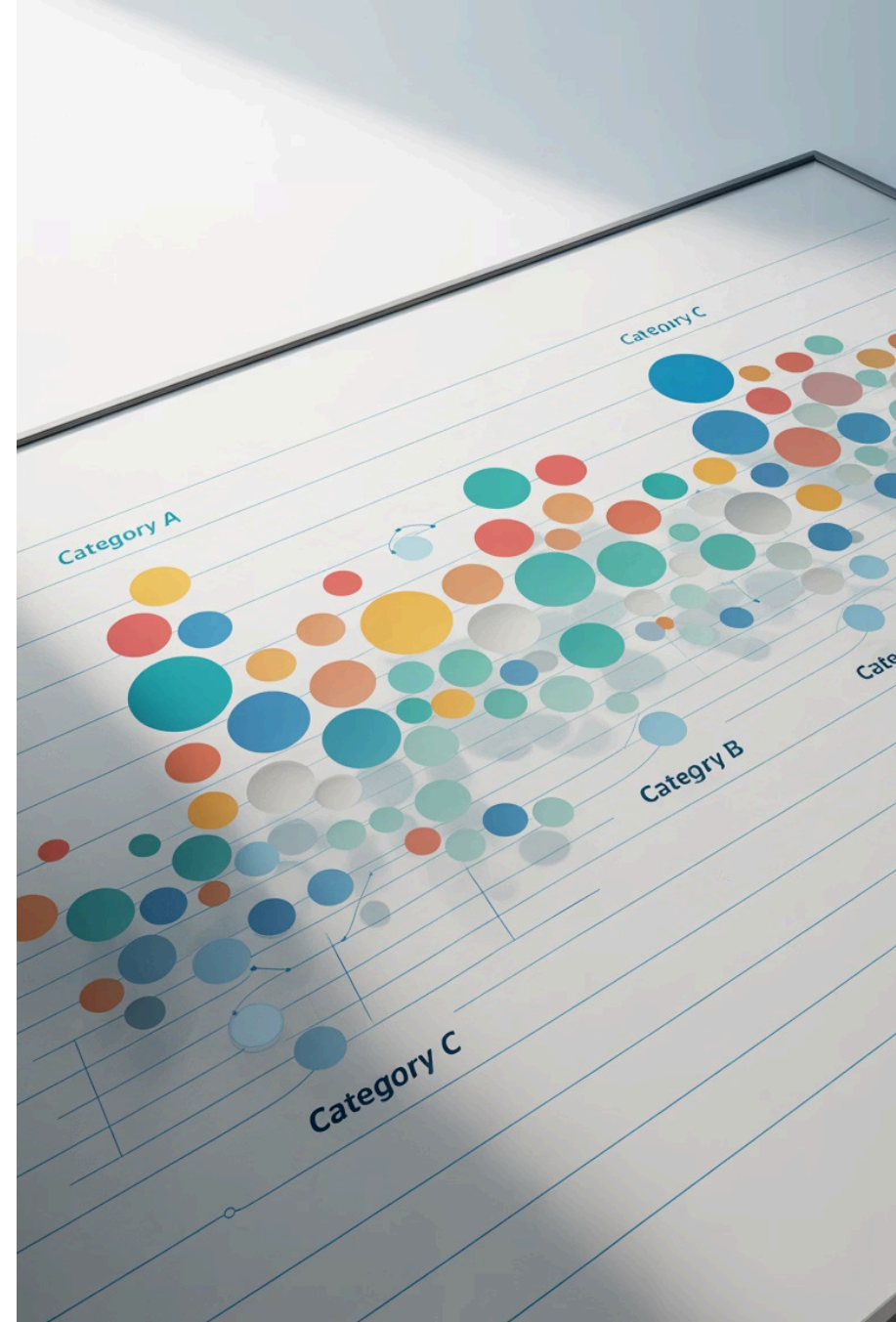
Main Challenges of Machine Learning

Insufficient Quantity of Training Data, **Non-representative Training Data**, Poor-Quality Data, Irrelevant Features, Overfitting the Training Data, Underfitting the Training Data, Stepping Back: Reviewing the Chapter So Far

Nonrepresentative Training Data: Overview

Importance of Representativeness:

- To **generalize well**, your **training data** must be **representative** of the **new cases** you want the **model** to **predict**.
- This applies whether you are using **instance-based learning** or **model-based learning**.



Example: Nonrepresentative Training Data

Initial Data:

In the earlier example, the dataset for training the **linear model** did not include countries with **GDP per capita** lower than **\$23,500** or higher than **\$62,500**.

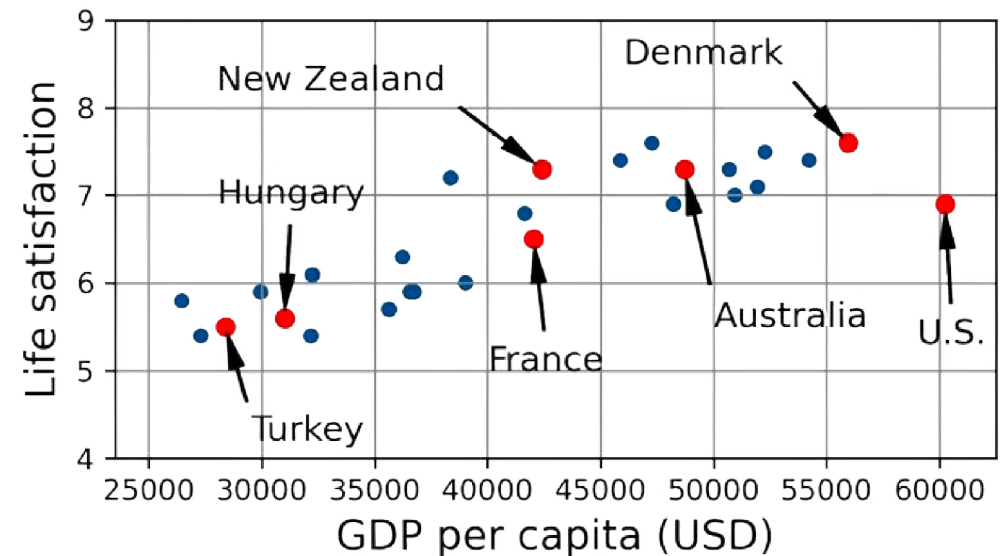


Figure 1-18. Do you see a trend here?

Effect of Missing Data:

Adding countries with a broader range of **GDP per capita** significantly alters the model.

Example: A more representative Training Data

The new model (solid line) is different from the original model (dotted line).

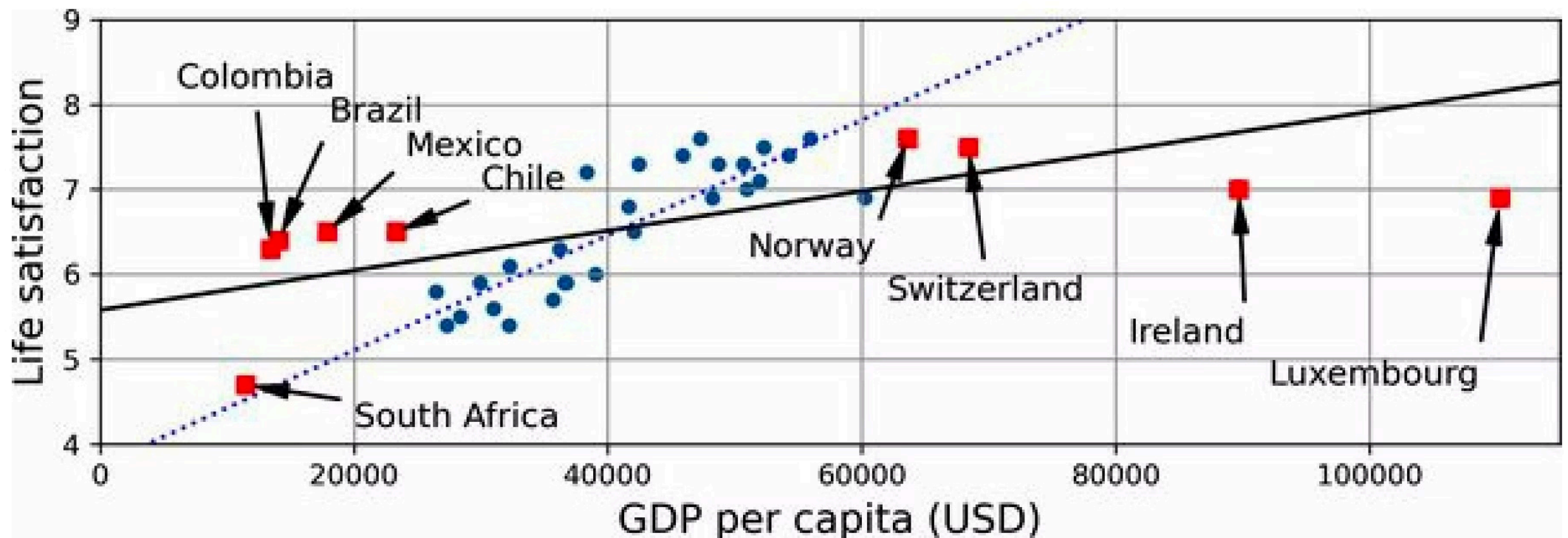


Figure 1-22. A more representative training sample

Outcome:

The new model reveals that:

- **Very rich countries are not necessarily happier** than moderately rich ones (in fact, they may be slightly unhappier).
- Some **poor countries** appear happier than many rich countries.



Sampling Bias



Problem

- **Non representative training sets** can lead to **models** that make **poor predictions**
 - Especially for **extreme cases** (very poor or very rich).
- The **dataset is biased**, leading to **inaccurate results**.



Sampling Bias

- **Sampling bias** occurs when the **sample** used to train the model is **flawed** or not **representative** of the **full population**.
- This results in **incorrect generalizations** and **reduced model accuracy**.

Examples of Sampling Bias

- 1936 US Presidential Election:
 - The Literary Digest conducted a large poll, but the sample was biased.
 - Sampling Method Flaws:
 - Poll recipients were drawn from **telephone directories**, magazine **subscribers**, and club **memberships**, favoring **wealthier people** (likely to vote **Republican**).
 - Nonresponse Bias:
 - **Less than 25% of people polled responded**, possibly excluding groups that didn't care about politics or had negative views about the poll.
 - **Outcome:** The poll predicted Landon would win, but Roosevelt won by a wide margin due to **sampling bias**.



Examples of Sampling Bias

- Funk Music Video Recognition:
 - Training Data Bias:
 - Searching “funk music” on YouTube may not yield a **representative sample** of all funk music videos, especially if the results are **biased** toward **popular artists** or **local genres** (e.g., **funk carioca** in Brazil).





Key Takeaways

Training Data Representativeness

Ensure the **training data reflects the full spectrum of cases** you want to **predict**, or your **model** may fail on **edge cases** (e.g., very poor or very rich countries).

Sampling Bias

Be mindful of **sampling bias** when **collecting** and **preparing training data**, as it can **lead to inaccurate predictions** and a **flawed model**.

Nonresponse Bias

Be cautious of **nonresponse bias** in **surveys** and **polls**, as it can **skew results** and lead to **false conclusions**.



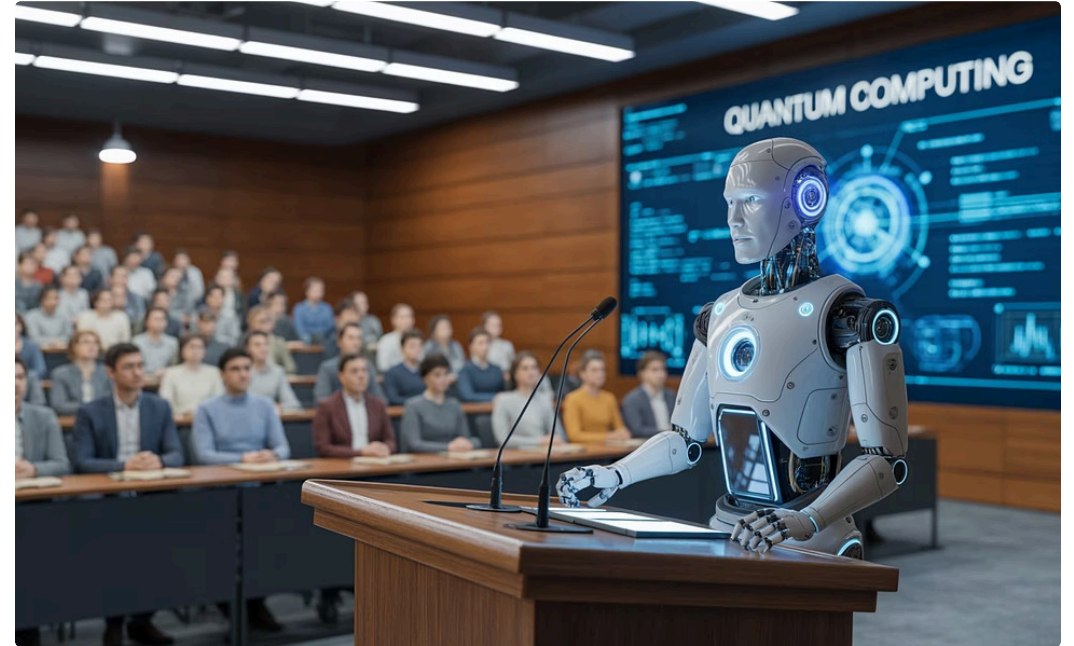
Next: Poor-Quality Data

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts